Ali Reza Ibrahimzada

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Research Interest

My research covers software engineering, machine learning, artificial intelligence, LLM agents, LLMs for code, and formal reasoning with the goal of modernizing and enhancing the reliability of large-scale software systems. To this end, I leverage program analysis and LLMs to automatically translate and validate repository-level code across different programming languages. So far, my work has focused on evaluating LLMs in code translation and investigating translation bugs [C5], source code decomposition for code translation [C4], and building end-to-end neuro-symbolic [C3] and agentic [P1] repository-level code translation and validation [C1, P2] tools.

Beside code translation, I have also enjoyed working on other problems in software engineering [C2, C6] and machine learning for bioinformatics [J1, J2].

Education

08/22 – 05/27 **Ph.D. Computer Science, University of Illinois Urbana-Champaign, IL, USA**
GPA: 4.00 / 4.00
Advisor: Reyhaneh Jabbarvand
Thesis: *Neuro-Symbolic Code Translation and Validation*

08/22 – 12/24 **M.Sc. Computer Science, University of Illinois Urbana-Champaign, IL, USA**
GPA: 4.00 / 4.00
Advisor: Reyhaneh Jabbarvand
Thesis: *Bridging the Gap between Testing and Debugging through Explainable Deep Oracles*

09/18 – 07/22 **B.Sc. Computer Engineering, Marmara University, Istanbul, Turkey**
GPA: 3.98 / 4.00
Advisor: Mehmet Kadir Baran
Thesis: *Depth Estimation of Stereo Images using Deep Learning*

Professional Experience

08/22 – Present **Research Assistant,** University of Illinois Urbana-Champaign, IL, USA
Advised by: Reyhaneh Jabbarvand
Topic: Code LLMs for Source Code Generation, Understanding and Translation

05/25 – 08/25 **Applied Scientist Intern,** Amazon Web Services, VA, USA
Hosted by: Brandon Paulsen and Joey Dodds
Topic: LLM Agents for Autonomous Translation Validation and Repair

05/24 – 08/24 **Research Intern,** IBM Research, NY, USA
Hosted by: Saurabh Sinha and Raju Pavuluri
Topic: End-to-End Repository-level Code Translation and Validation

05/23 – 08/23 **Research Intern,** IBM Research, NY, USA
Hosted by: Saurabh Sinha and Raju Pavuluri
Topic: Evaluating LLMs in Code Translation and Bug Analysis

05/21 – 08/22 **Research Intern,** University of Illinois Urbana-Champaign, IL, USA
Hosted by: Reyhaneh Jabbarvand
Topic: Neural and Interpretable Test Oracles using Deep Learning

06/19 – 02/22 **Research Intern,** Istanbul Technical University, Istanbul, Turkey
Hosted by: Ali Cakmak
Topic: Machine Learning for Bioinformatics

Research Publications

Preprints

[P1] **A. R. Ibrahimzada**, B. Paulsen, D. Kroening, and R. Jabbarvand, “Recodeagent: A multi-agent workflow for language-agnostic translation and validation of large-scale repositories,” 2026. URL: https://arxiv.org/abs/2604.07341.

[P2] K. Ke, **A. R. Ibrahimzada**, R. Pan, S. Sinha, and R. Jabbarvand, “Advancing automated in-isolation validation in repository-level code translation,” 2025. URL: https://arxiv.org/abs/2511.21878.

Conference Proceedings

[C1] **A. R. Ibrahimzada**, B. Paulsen, R. Jabbarvand, J. Dodds, and D. Kroening, “Matchfixagent: Language-agnostic autonomous repository-level code translation validation and repair,” in *International Conference on Machine Learning*, PMLR, 2026. URL: https://openreview.net/forum?id=MuyXpH3GL1.

[C2] **A. R. Ibrahimzada**, Y. Chen, R. Rong, and R. Jabbarvand, “Challenging bug prediction and repair models with synthetic bugs,” in *2025 IEEE International Conference on Source Code Analysis & Manipulation (SCAM)*, 2025, pp. 133–144. DOI: 10.1109/SCAM67354.2025.00021.

[C3] **A. R. Ibrahimzada**, K. Ke, M. Pawagi, M. S. Abid, R. Pan, S. Sinha, and R. Jabbarvand, “Alphatrans: A neuro-symbolic compositional approach for repository-level code translation and validation,” FSE, vol. 2, New York, NY, USA: Association for Computing Machinery, Jun. 2025. DOI: 10.1145/3729379.

[C4] **A. R. Ibrahimzada**, “Program decomposition and translation with static analysis,” in *Proceedings of the 2024 IEEE/ACM 46th International Conference on Software Engineering: Companion Proceedings*, ser. ICSE-Companion ’24, Lisbon, Portugal: Association for Computing Machinery, 2024, pp. 453–455, ISBN: 9798400705021. DOI: 10.1145/3639478.3641226.

[C5] R. Pan*, **A. R. Ibrahimzada***, R. Krishna, D. Sankar, L. P. Wassi, M. Merler, B. Sobolev, R. Pavuluri, S. Sinha, and R. Jabbarvand, “Lost in translation: A study of bugs introduced by large language models while translating code,” in *Proceedings of the IEEE/ACM 46th International Conference on Software Engineering*, ser. ICSE ’24, Lisbon, Portugal: Association for Computing Machinery, 2024, ISBN: 9798400702174. DOI: 10.1145/3597503.3639226.

[C6] **A. R. Ibrahimzada**, Y. Varli, D. Tekinoglu, and R. Jabbarvand, “Perfect is the enemy of test oracle,” in *Proceedings of the 30th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering*, ser. ESEC/FSE 2022, Singapore, Singapore: Association for Computing Machinery, 2022, pp. 70–81, ISBN: 9781450394130. DOI: 10.1145/3540250.3549086.

Journal Articles

[J1] A. Cakmak, H. Ayaz, S. Arıkan, **A. R. Ibrahimzada**, Ş. Demirkol, D. Sönmez, M. T. Hakan, S. T. Sürmen, C. Horozoğlu, M. B. Doğan, Ö. Küçükhüseyin, C. Cacına, B. Kıran, Ü. Zeybek, M. Baysan, and İ. Yaylım, “Predicting the predisposition to colorectal cancer based on snp profiles of immune phenotypes using supervised learning models,” *Medical & Biological Engineering & Computing*, vol. 61, no. 1, pp. 243–258, Jan. 2023, ISSN: 1741-0444. DOI: 10.1007/s11517-022-02707-9.

[J2] G. N. Sohsah, **A. R. Ibrahimzada**, H. Ayaz, and A. Cakmak, “Scalable classification of organisms into a taxonomy using hierarchical supervised learners,” *Journal of Bioinformatics and Computational Biology*, vol. 18, no. 05, p. 2 050 026, 2020, PMID: 33125294. DOI: 10.1142/S0219720020500262. eprint: https://doi.org/10.1142/S0219720020500262.

Thesis

[T1] **A. R. Ibrahimzada**, *Bridging the gap between testing and debugging through explainable deep oracles*, 2024.

Awards

2024	**Nomination for the Two Sigma PhD Fellowship**, University of Illinois Urbana-Champaign
	3rd in Student Research Competition @ ICSE'24, ACM
	Travel Grant to attend ICSE'24 in Lisbon, Portugal, NSF
2022	**Travel Grant to attend ESEC/FSE'22 in Singapore, Singapore**, ACM
	Valedictorian of Class of 2022, Marmara University
	Ray Ozzie Computer Science Fellowship, University of Illinois Urbana-Champaign
	Best Senior Graduation Project of the Year, Marmara University
2019	**Academic Achievement Scholarship**, Marmara University
2017	**Valedictorian of High School**, KEN
2015	**Travel Grant to attend QUEST'15 in Lucknow, India**, KEN

Talks

SCAM'25	Challenging Bug Prediction and Repair Models with Synthetic Bugs, Auckland, New Zealand
FSE'25	AlphaTrans: A Neuro-Symbolic Compositional Approach for Repository-Level Code Translation and Validation, Trondheim, Norway
ICSE'24	Lost in Translation: A Study of Bugs Introduced by Large Language Models while Translating Code, Lisbon, Portugal
FSE'22	Perfect Is the Enemy of Test Oracle, Singapore, Singapore

Skill Stack

Programming Languages	Python, Java, JavaScript, Rust, C, OCaml
Common	Docker, Vim, Git, SQL
Machine Learning	PyTorch, TensorFlow, vLLM, Codex, Claude Code

Service

Program Committee	NeurIPS'26 (Sydney, Australia), ICML'26 (Seoul, South Korea), ICLR'26 (Rio de Janeiro, Brazil), ICLR'25 (Singapore, Singapore), MSR'25 (Ottawa, Canada), MSR'24 (Lisbon, Portugal)
Artifact Evaluation	ICSE'25 (Ottawam Canada)
Organizing	ReCode@ICSE'26 (Rio de Janeiro, Brazil)

Teaching Experience

Fall 2025	CS 421 - Programming Languages and Compilers w/ Elsa Gunter
Spring 2025	CS 447 - Natural Language Processing w/ Julia Hockenmaier

References

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